preface

The late 1990s found me at IIT Kharagpur in India, where I was pursuing my master's degree and simultaneously immersing myself in the fascinating realm of the Java language (version 1, to be precise!). It was a solitary journey, although I enjoyed using an object-oriented language for the first time, each code compilation failure signifying a duel between my terminal and me. There were no fancy IDEs in those days; we typed in a vi editor and then compiled. (There was a rudimentary IDE called Java WorkShop from Sun Microsystems, but it was not without wrinkles and bugs.) And unlike in the present era, which is replete with resources like Google, Stack Overflow, and GitHub, there were no digital avenues to probe for solutions or discover if other programmers were encountering similar challenges. Then my best friend Amar, who was studying at IISc, Bangalore, introduced me to a new entity in the search arena: Google.

To a naïve young student, the concept of reaching out to a virtual community of coders grappling with similar issues was a novelty. So, I decided to explore this new avenue. I remember being intrigued by the simplicity of Google's design, its non-eyesore white background dotted with primary colors, as I navigated pages in search of answers. The early version of the search engine was far from the sophisticated tool we know today, but it was an encouraging beacon during those challenging times. (Anyone who has experienced Java 1.x can understand the trials involved.) That moment marked a significant turning point in my journey.

While Google's search capabilities were revolutionary for many of us, its adoption within client settings and organizations was somewhat slower. Projects I was involved in at that time had databases serving as the backbone for search applications. While functional, these setups fell considerably short of what we now recognize as the capabilities of modern search engines. The search mechanisms were clunky and unwieldy,